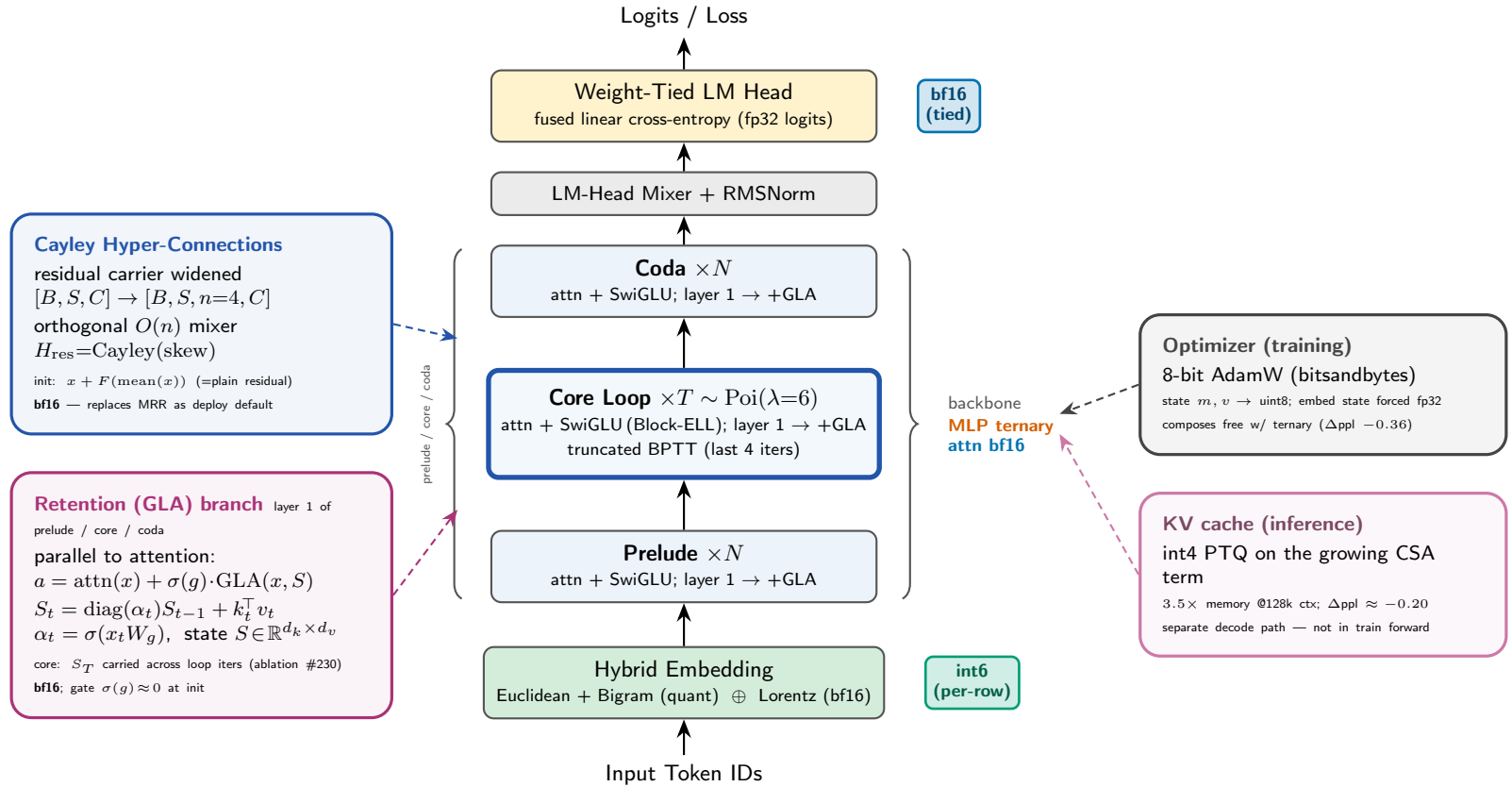


MORPH — Deployable Stack



Precision legend (text label is authoritative; color is secondary):

int6

embeddings

ternary

backbone MLP

bf16

attn /

norms / Lorentz / HC / GLA

int4

KV cache

8-bit

optimizer state

Validated evidence (15k, b4, vs bf16 baseline **ppl 39.46**): ternary backbone $\rightarrow$ 42.07 (+6.6% for 22.9% of params $\rightarrow \{-1, 0, 1\}$); int6 embeds $\approx$ free (72% of params); attn left bf16 (all-ternary attn = 44.92 $\gg$); int4 KV $\Delta \text{ppl} \approx 0$; 8-bit AdamW interaction -0.36 .